

# 2020: The State of Accessibility & Inclusion

March 2020 // Dennie Declercq // Finland

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# Who am I?

- Dennie Declercq
- Volunteer developer!
- Autism
- Microsoft MVP
- DDSOft (nonprofit)
- #DreamingIsBelieving
- Positive thinker



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# Content

- What was the start promise of Inclusion?
- 2020: Inclusion for PWD: Where are we now?
- Accessible software / app advices
- Inclusive advices beyond software
- Use Cases Best Practices on Inclusion / Accessibility
- And right now, it's up to you!
- Inclusion is not only about PWD anymore
- Conclusion: Inclusion helps!
- Thanks | Q&A

# What was the start promise of Inclusion?

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# The first discussion of inclusion



- People with disabilities
  - Comparing to other people
- Later on: inclusion expanded



# The need of inclusion

- Until 2000:
  - PWD separated away
  - Specialized clinics and institutions
  - No place for them



# 2000: The beginning of inclusion



- Early 2000's
  - Not the way to go
  - Change is needed
- Prof. Bob Shallock
- Mr. Arnold Van Gennep
- PWD deserve:
  - QoL
  - Life as other people



# Inclusion in theory in very short words

- PWD:
  - Same living environment, workplaces, hobby , sports clubs
  - Same QoL
  - Schalock's Domains of Wellbeing





# Inclusion in the human words



- A neighbor with disabilities
- A person with special needs in the grocery store
- A blind person ordering his / her own food
- A child with autism having friends without autism
- More explanation if needed
- Which assistance is not wanted
- A co worker with special needs

# Inclusion in developers words



AI that helps  
PWD making  
choices



A timer



Text-to-speech  
for learning  
Disabilities



The option to  
toggle dark  
screen



The choice  
between  
landscape office  
or at home



Engineering  
possibilities that  
helps everyone



....



# 2020: Inclusion for PWD: Where are we now?

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# Where do people with disabilities live?

- Inclusive living projects
- Consider variety
- Not everyone looks like a stranger....
- More people with special needs in Public Transport
- PWD have social activities and a network



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# What's the current state of digital assistance?



- App / Software vendors making apps for PWD
- Some apps limit PWD
- Some apps giving more to PWD
- Most of the time too expensive
- Need permission parent / caregiver
- No good balance yet

# Let's talk about trust

- At This moment the biggest struggle
- Way to much caregivers:
  - Too little trust
  - Too little knowledge
  - In digital software, software vendors and digital technology
- Too many people biased:
  - P.w. learning disabilities can't...
- The step to pay.... Is too big





# Are people with disabilities connected?



- The most PWD are connected
- Via internet to other people, friends and family
- Most caring center have WIFI
- This is really good!

# Do people with disabilities have a mobile phone?

- A lot of blind people have an iPhone
- A lot of p. w. learning disability:
  - Smartphone
    - Including some kind of assistive technology
- Conclusion: opportunity is here!

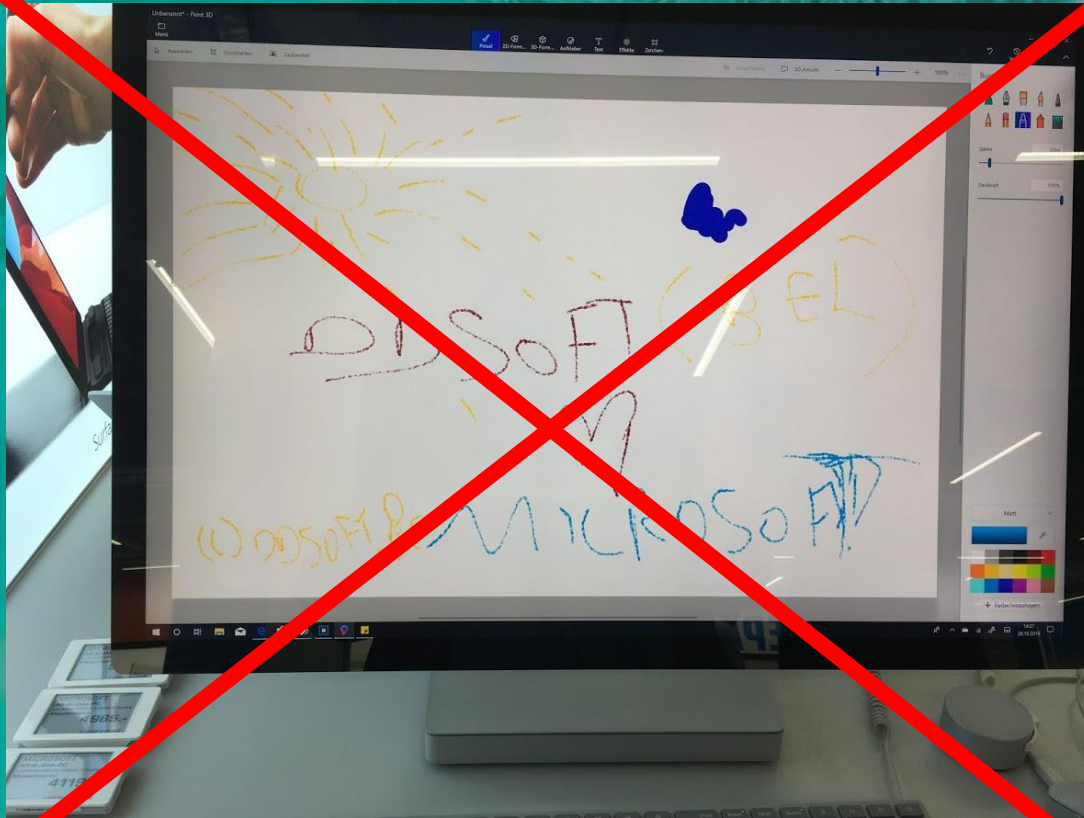


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# What about hardware and software versions?



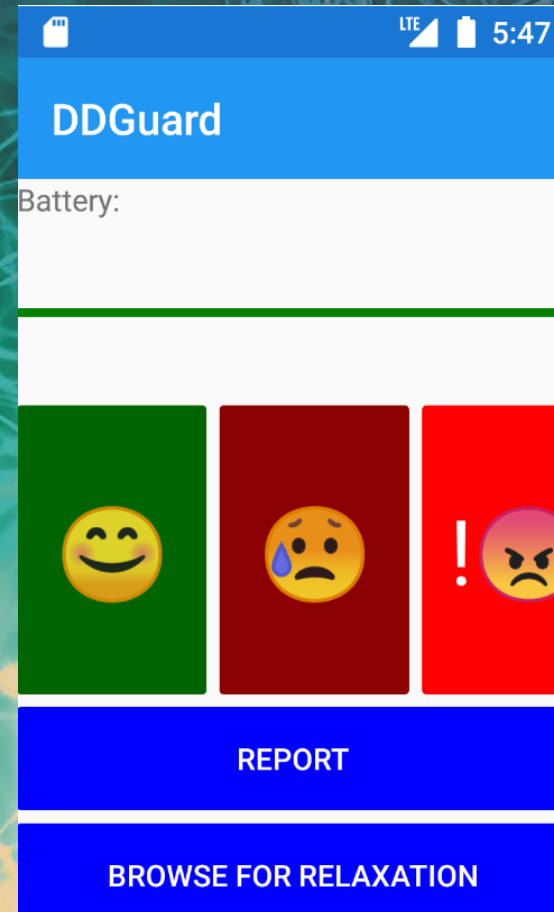
- Mostly cheaper versions
- Outdated OS
- Deprecated software
- Low hardware specs

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# Can mobile apps really help QoL?

- Yes!
- Saw a lot of use cases.
- Increase of QoL is real thing!
- More interdependency
- Less dependency

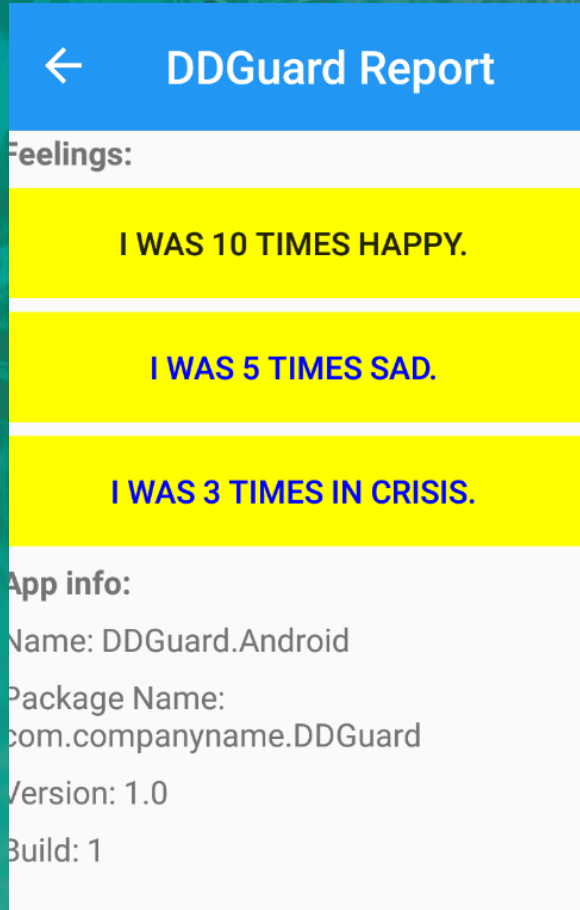


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# But that apps need to work!



- People losing trust very fast if:
  - Assistive tech fails
  - Assistive apps require more computing
- If trust is gone, the use of the app / smartphone.. Is gone

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# Accessible Software / App advices

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# Layout patterns

- Simplify The layout
- **Limited** content
- Direction
- Use **Accent** Colors
- Option Toggle **Dark Theme**

# Content Patterns

- Target own language
- Use an easy language

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# Text-To-Speech

- **TTS** (Text-To-Speech) for p. w. **Learning Disabilities**
- **TTS != Voice Over, Narrator or Read Aloud**

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# Pictures and Pictographs

- What means the following?
- 酢飯
- The answer





# Color By Function

I speak

I open another  
page

I navigate

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# Inclusive advices beyond software

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# Hiring and people with disabilities

- Advantages of Hiring:
  - P.w. Autism
    - Problem solving
    - Good developers
    - Good testers
  - A blind person
    - Coding (with adjustments)
  - Premises Wheelchair accessible?
    - Working from home?
  - Hearing difficulties
    - Communication system, Emergency situations



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# Making an inclusive event experience



- Reserved seating
- Deaf translators
- **Attendees:** leave open place in your discussion group!

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**Can accessible software /  
apps / services solve real  
world problems of PWD**

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# Difficulties in analog world



NAVIGATION IN  
WORLD / CITY



TIME &  
PLANNING



SHOPPING



BUDGET



HOUSE  
CLEANING



CLOTHING

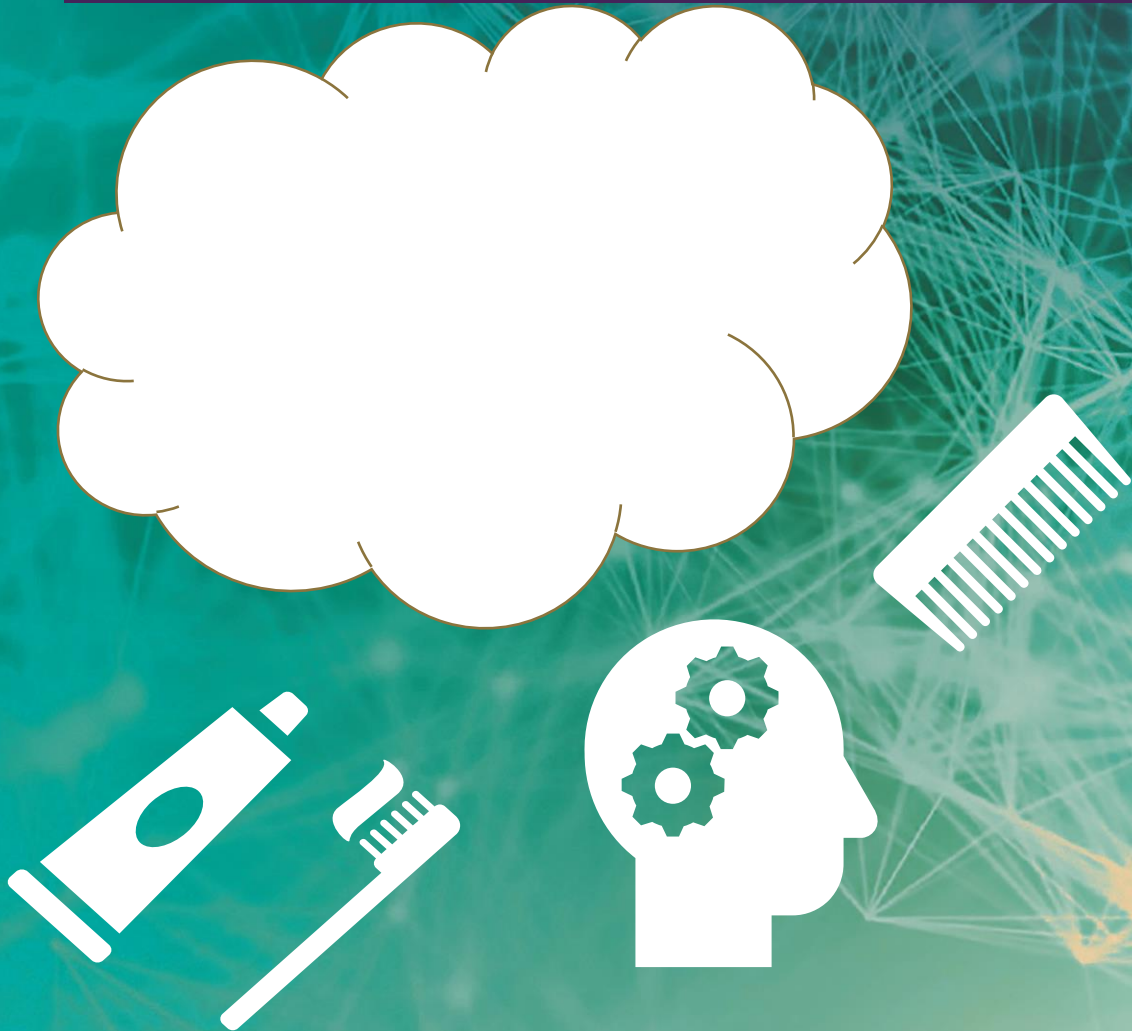


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# Opportunities

- Guiding applications
- Cloud & AI



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# Use Cases Best Practices on Inclusion / Accessibility

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# Microsoft Inclusive Hiring



## Inclusive Hiring

<https://www.microsoft.com/en-us/diversity/inside-microsoft/cross-disability/hiring.aspx?SilentAuth=1>

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# Microsoft Seeing AI

## Seeing AI in new languages

A free app that narrates the world around you, now available in Dutch, German, French, Japanese and Spanish.

[Learn more about Seeing AI >](#)

[Download on the App Store >](#)

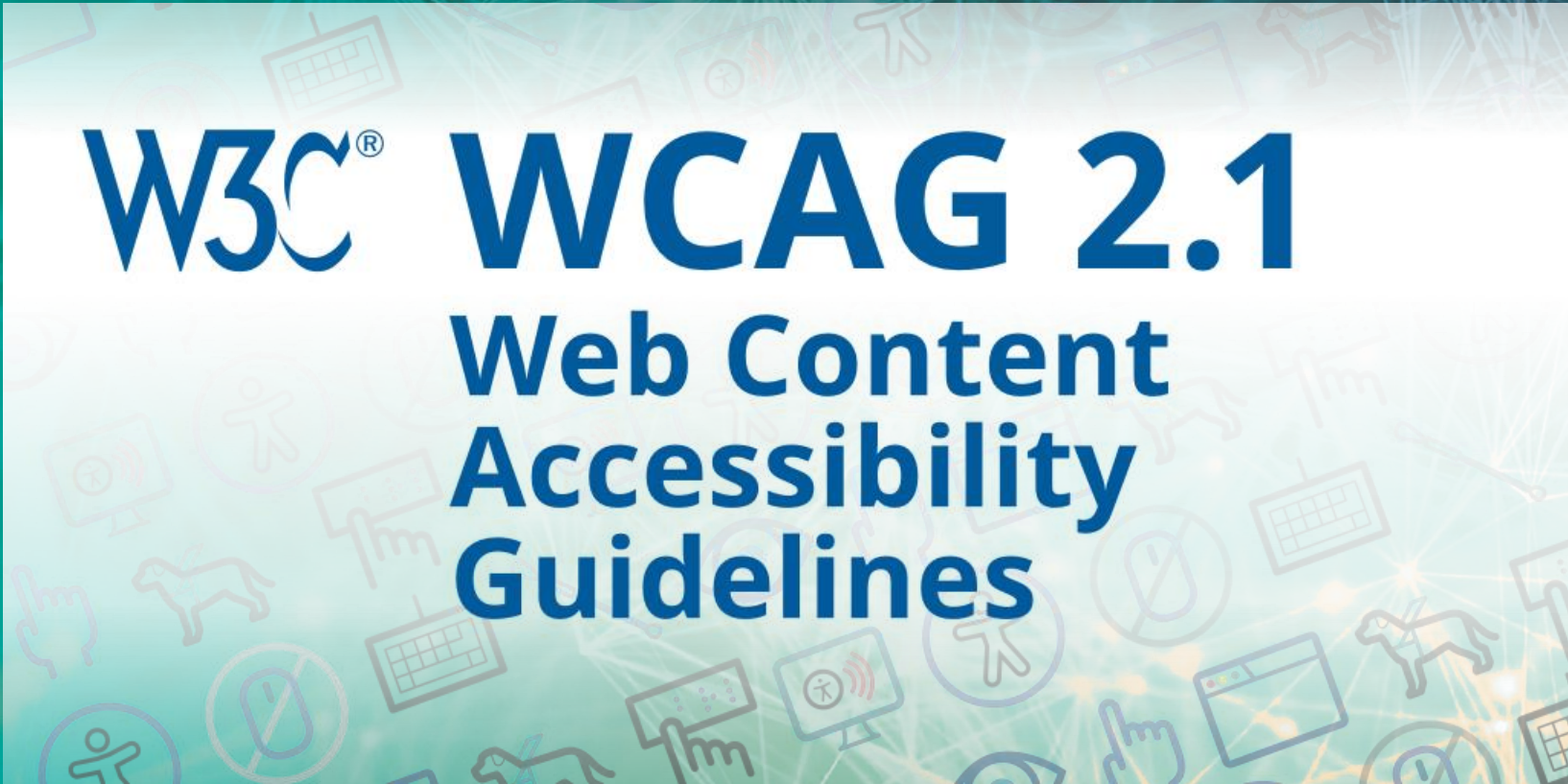
<https://www.microsoft.com/en-us/ai/seeing-ai>

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# Web Content Accessibility Guidelines (WCAG) 2.0

The logo for WCAG 2.1 is centered on a white rectangular background. The background of the entire slide is a teal-to-blue gradient with a network of glowing orange and yellow lines. Faint, semi-transparent icons representing various accessibility needs (like a person in a wheelchair, a hand with a pointer, a keyboard, a dog, and a speech bubble) are scattered across the white background.

## W3C<sup>®</sup> WCAG 2.1

### Web Content Accessibility Guidelines

<https://www.w3.org/TR/WCAG20/>

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# UNFORTUNATELY: I can't add a long list of app vendors



BlindSquare

Sunday, September 16, 12



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**And right now, It's up to you!**

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# Start a customer research project

- Research about:
  - Accessibility & Inclusion
  - How can you improve QoL?



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# Are all of your features used?

- Features nobody use?
- It's more confusing
- Option to leave features?



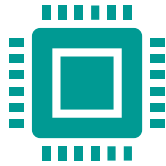
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# Did you hear any complains of PWD?



**Feedback of PWD**



**Missing the use of your software?**



**Release an update**

Accessibility & Inclusion minded?



**Guidelines learned in this talk.**

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# Let's talking about hiring and co-workers







# Inclusion is not only about PWD anymore

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# Inclusion and Mental Health



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# Inclusion and Woman in Tech



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# Inclusion and migration & people of color



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# Inclusion and gender diversity



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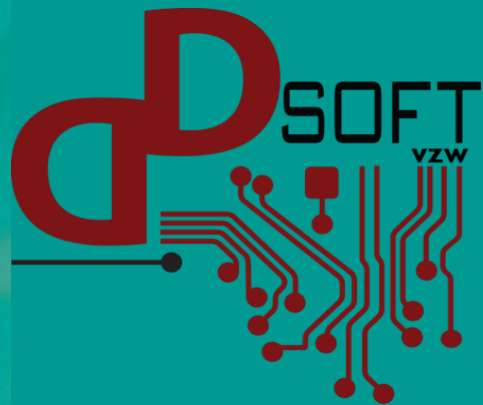
# Conclusion: Inclusion helps!

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# Thanks || Q&A

- Dennie Declercq
- @DennieDeclercq
- #DreamingIsBelieving
- [dennie@ddsoft.be](mailto:dennie@ddsoft.be)



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